6. Ratio and Proportion

Proportion : **Equality of two ratios is called a proportion.** $\frac{a}{b}=\frac{c}{d}$ **is a proportion. This is written as** $a:b=c:d$

Here a,b,c,d **are called proportionals.**

Properties of proportions.

If $a:b=c:d$ **, then**
❶ $\frac{a+b}{b}=\frac{c+d}{d}$
❷ $\frac{a-b}{b}=\frac{c-d}{d}$
❸ $\frac{a-b}{a-b}=\frac{c+d}{c-d}$

Let $\frac{a}{b}=\frac{c}{d}=k$

$\Rightarrow a=kb$ **and** $c=kd$

❶ $\frac{a+b}{b}=\frac{kb+b}{b}=\frac{b(k+1)}{b}=k+1$

$\frac{c+d}{d}=\frac{kd+d}{d}=\frac{d(k+1)}{d}=k+1$

Hence $\frac{a+b}{b}=\frac{c+d}{d}$

❷ $\frac{a-b}{b}=\frac{kb-b}{b}=\frac{b(k-1)}{b}=k-1$

$\frac{c-d}{d}=\frac{kd-d}{d}=\frac{d(k-1)}{d}=k-1$

Hence $\frac{a-b}{b}=\frac{c-d}{d}$

❸ $\frac{a+b}{a-b}=\frac{kb+b}{kb-b}=\frac{(k+1)b}{(k-1)b}=\frac{k+1}{k-1}$

$\frac{c+d}{c-d}=\frac{kd+d}{kd-d}=\frac{(k+1)d}{(k-1)d}=\frac{k+1}{k-1}$

Hence $\frac{a+b}{a-b}=\frac{c+d}{c-d}$

If $\frac{a}{b}=\frac{c}{d}$, **then each ratio is equal to** $\frac{ma+nc}{mb+nd}$

i.e, If $\frac{a}{b}=\frac{c}{d}$, **then** $\frac{a}{b}=\frac{c}{d}=\frac{ma+nc}{mb+nd}$

Let $\frac{a}{b}=\frac{c}{d}=k$

$a=kb$ **and** $c=kd$

$$\frac{ma+nc}{mb+nd}=\frac{kmb+knd}{mb+nd}=\frac{k(mb+nd)}{(mb+nd)}=k$$

Hence $\frac{a}{b}=\frac{c}{d}=\frac{ma+nc}{mb+nd}$

This is a very useful result in solving problems.

Example 1

If $\frac{4a+b}{2a+b}=7$, **find the values of (a)** $\frac{5a+b}{5a-b}$ **(b)** $\frac{b^2-a^2}{a^2+b^2}$

$$\frac{4a+b}{2a+b}=7$$

$$4a+b=14a+7b$$

$$10a=-6b$$

$$5a=-3b$$

$$a=-\frac{3b}{5}$$

(a) $\frac{5a+b}{5a-b}=\frac{-3b+b}{-3b-b}=\frac{-2b}{-4b}=\frac{1}{2}$

(b) $\frac{b^2-a^2}{b^2+a^2}=\frac{b^2-\frac{9b^2}{25}}{b^2+\frac{9b^2}{25}}=\frac{16}{34}=\frac{8}{17}$

Example 2

Solve the equations

$$2x - 3y = 0$$
$$3x + 4y = 51$$

$2x - 3y = 0, \qquad 2x = 3y$

$$\frac{x}{3} = \frac{y}{2}$$

Let $\frac{x}{3} = \frac{y}{2} = k$

Then $k = \frac{x}{3} = \frac{y}{2} = \frac{3x+4y}{3\times 3+4\times 2} = \frac{51}{17} = 3$

$$\frac{x}{3} = \frac{y}{2} = 3$$

$$\left.\begin{aligned} x &= 9 \\ y &= 6 \end{aligned}\right\}$$

Example 3

If $\frac{x}{y} = \frac{a}{b}$, **show that** $\frac{2x+3y}{2x-3y} = \frac{2a+3b}{2a-3b}$

$\frac{x}{y} = \frac{a}{b}$ **implies that** $\frac{x}{a} = \frac{y}{b}$

Let $k = \frac{x}{a} = \frac{y}{b}$

Now $k = \frac{x}{a} = \frac{y}{b} = \frac{2x+3y}{2a+3b}$

$$k = \frac{x}{a} = \frac{y}{b} = \frac{2x-3y}{2a-3b}$$

$$\frac{2x+3y}{2a+3b} = \frac{2x-3y}{2a-3b}$$

$$\frac{2x+3y}{2x-3y} = \frac{2a+3b}{2a-3b}$$

Alternate Method

Let $\frac{x}{y} = \frac{a}{b} = k$

$x = ky$ **and** $a = kb$

$$\frac{2x+3y}{2x-3y}=\frac{2ky+3y}{2ky-3y}=\frac{2k+3}{2k-3}$$

$$\frac{2a+3b}{2a-3b}=\frac{2kb+3b}{2kb-3b}=\frac{2k+3}{2k-3}$$

Hence $\frac{2x+3y}{2x-3y}=\frac{2a+3b}{2a-3b}$

Example 4

If $(4a+b)(4c-7d)=(4a-7b)(4c+d)$

Show that $a:b=c:d$

$$(4a+b)(4c-7d)=(4a-7b)(4c+d)$$

$$\frac{4a+b}{4c+d}=\frac{4a-7b}{4c-7d}=k(say)$$

$$k=\frac{4a+b}{4c+d}=\frac{4a-7b}{4c-7d}=\frac{(4a+b)-(4a-7b)}{(4c+d)-(4c-7d)}=\frac{8b}{8d}=\frac{b}{d}$$

$$k=\frac{4a+b}{4c+d}=\frac{4a-7b}{4c-7d}$$

$$k=\frac{7(4a+b)}{7(4c+d)}=\frac{4a-7b}{4c-7d}=\frac{7(4a+b)+(4a-7b)}{7(4c+d)+(4c-7d)}=\frac{32a}{32c}=\frac{a}{c}$$

Her $\frac{a}{c}=\frac{b}{d}$

$$\Rightarrow \frac{a}{b}=\frac{c}{d}$$

Example 4 (Alternate Method)

If $(4a+7b)(4c-7d)=(4a-7b)(4c+7d)$

Show that $a:b=c:d$

Let $\frac{a}{b}=m$ **and** $\frac{c}{d}=n$

$a=mb$ **and** $c=nd$

Substituting $a=mb$ **and** $c=nd$ **in the equation**

$$(4a+7b)(4c-7d)=(4a-7b)(4c+7d)$$

We get; $(4mb+7b)(4nd-7d) = (4mb-7b)(4nd+7d)$

$$bd(4m+7)(4n-7) = bd(4m-7)(4n+7)$$

$$b, d \neq 0$$

$$\therefore (4m+7)(4n-7) = (4m-7)(4n+7)$$

$$16mn+28n-28m-49 = 16mn+28m-28n-49$$

$$28n-28m = 28m-28n$$

$$n-m = m-n$$

$$2m = 2n$$

$$m = n$$

Hence $a : b = c : d$

Example 5

If $x = \dfrac{6ab}{a+b}$, **find the value of** $\dfrac{x+3a}{x-3a} + \dfrac{x+3b}{x-3b}$

$$x = \frac{6ab}{a+b} \Rightarrow \frac{x}{2b} = \frac{3a}{a+b}$$

$$\frac{x}{2b} = \frac{3a}{a+b} = \frac{x+3a}{2b+(a+b)} = \frac{x-3a}{2b-(a+b)}$$

$$\frac{x+3a}{3b+a} = \frac{x-3a}{b-a}$$

$$\frac{x+3a}{x-3a} = \frac{3b+a}{b-a} \quad \text{———— (1)}$$

$$x = \frac{6ab}{a+b} \Rightarrow \frac{x}{2a} = \frac{3b}{a+b}$$

$$\frac{x}{2a} = \frac{3b}{a+b} = \frac{x+3b}{2a+(a+b)} = \frac{x-3b}{2a-(a+b)}$$

$$\frac{x+3b}{3a+b} = \frac{x-3b}{a-b}$$

$$\frac{x+3b}{x-3b} = \frac{3a+b}{a-b} \quad \text{———— (2)}$$

From (1) and (2)

$$\frac{x+3a}{x-3a} + \frac{x+3b}{x-3b} = \frac{3b+a}{b-a} + \frac{3a+b}{a-b}$$

$$= \frac{(3b+a)-(3a+b)}{b-a}$$

$$= \frac{2(b-a)}{b-a}$$

$$= 2$$

Exercise 6

1 (a) If $a:b=5:3$ and $b:c=4:5$ find $a:b:c$

(b) If $x:y=3:4$ and $x:z=4:5$, find $x:y:z$

2 If $x:y=7:5$ find $5x-2y:3x+2y$

3 If $3x+5y:5x+12y=11:12$, find $x:y$

4 If, $5a^2-ab:2ab-b^2=6:1$, find $a:b$

5 If $a:b=c:d$, prove that

(a) $(2a+3b):(2c+3d)=(2a-3b):(2c-3d)$

(b) $(3a+5b):(3a-5b)=(3c+5d):(3c-5d)$

6 If $x=\dfrac{2ab}{a+b}$, find the value of $\dfrac{x+a}{x-a}+\dfrac{x+b}{x-b}$

7 If $x=\dfrac{10ab}{a+b}$, find the value of $\dfrac{x+5a}{x-5a}+\dfrac{x+5b}{x-5b}$

8 If $(2a+3b)(2c-3d)=(2a-3b)(2c+3d)$, show that $a:b=c:d$

9 If $(3a+6b-c-2d)(3a-6b+c-2d)=(3a+6b+c+2d)(3a-6b-c+2d)$, show that $a:b=c:d$

10 Solve the following equations using the properties of proportion

(a) $\dfrac{x^2+1}{2x}=\dfrac{5}{4}$ (b) $\dfrac{x^3+3x}{3x^2+1}=\dfrac{341}{91}$